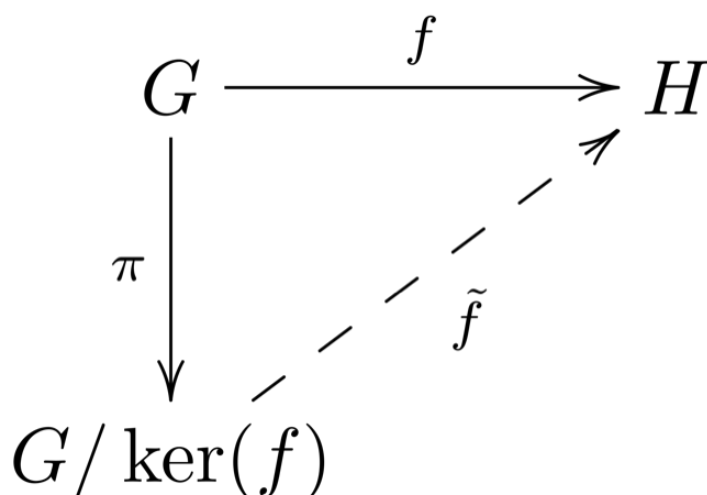


Chapter 5. Factor groups

5.1 Normal subgroups



In this video, we will talk about a very important type of subgroups, called normal subgroups.

What are normal subgroups?

Definition 5.1. We say that H is a **normal subgroup** of G , or that H is normal in G , if $gH = Hg$ for all $g \in G$. In such a case, we write $H \triangleleft G$.

Examples 5.2.

1. Let G be any group. Then $\{1\} \triangleleft G$ and $G \triangleleft G$ since $g\{1\} = \{g\} = \{1\}g$, $gG = G = Gg$, for all $g \in G$. The centre $Z(G)$ of G is always a normal subgroup of G .
2. Every subgroup of an abelian group is normal.
3. Let $G = S_3 = \{\varepsilon, \sigma, \sigma^2, \tau, \tau\sigma, \tau\sigma^2\}$, where $\sigma^3 = \tau^2 = \varepsilon$ and $\sigma\tau\sigma = \tau$. Write $H = \{\varepsilon, \tau\}$ and $K = \{\varepsilon, \sigma, \sigma^2\}$. Then one can check that $K \triangleleft S_3$ but H is not normal in S_3 . In fact:

$$\tau = (1 \ 2)$$

$$\sigma = (1 \ 2 \ 3)$$

Proceeding like in Example 4.2, you can easily check that H is not normal in G .

We prove here that $K = \{\epsilon, \sigma, \sigma^2\} \triangleleft S_3$.

Using the coset properties we have that

$$\sigma K = K = K\sigma$$

$$\sigma^2 K = K = K\sigma^2$$

$$\tau K \stackrel{?}{=} K\tau$$

$$\underline{\tau K} = \{\tau, \tau\sigma, \tau\sigma^2\}$$

$$K\tau = \{\tau, \underline{\sigma\tau}, \underline{\sigma^2\tau}\} \stackrel{(*)}{=} \{\tau, \tau\sigma^2, \tau\sigma\} = \underline{\tau K}$$

$$(*) \begin{cases} \sigma\tau\sigma = \tau \Rightarrow \sigma\underline{\tau\sigma^3} = \tau\sigma^2 \Rightarrow \sigma\tau = \underline{\tau\sigma^2} \\ \sigma\tau\sigma = \tau \Rightarrow \sigma^3\underline{\tau\sigma} = \sigma^2\tau \Rightarrow \underline{\tau\sigma} = \sigma^2\tau \end{cases}$$

Exercise

$$(\tau\sigma)K = K(\tau\sigma)$$

$$(\tau\sigma^2)K = K(\tau\sigma^2)$$

How can we check that a subgroup is normal?

THEOREM 5.3. NORMALITY TEST. *The following conditions are equivalent for a subgroup H of a group G .*

- (1) H is normal in G .
- (2) $gHg^{-1} \subseteq H$ for all $g \in G$.

Proof (1) $\Rightarrow$ (2)

Suppose that $H \triangleleft G$. For $g \in G$ we need to check that $gHg^{-1} \stackrel{?}{=} H$.

Take $x \in gHg^{-1}$, then $x = \underline{ghg^{-1}}$ for some $h \in H$.

$$\underline{gh} \in gH \stackrel{\substack{\uparrow \\ H \text{ is normal}}}{=} \underline{Hg} \Rightarrow gh = \underline{h'g} \text{ for some } h' \in H.$$

$$\begin{aligned} \underline{x} &= ghg^{-1} = (gh)g^{-1} = (h'g)g^{-1} = h'(gg^{-1}) \\ &= h' \in H \end{aligned}$$

② $\Rightarrow$ ① : Suppose that $gHg^{-1} \subseteq H$
for all $g \in G$.

Take $g \in G$ and let us show $gH \stackrel{?}{=} Hg$.

By ② we have $g^{-1}H(g^{-1})^{-1} \subseteq H$,
that is, $g^{-1}Hg \subseteq H$.

$$\begin{aligned} \underline{H} &= (gg^{-1})H(gg^{-1}) = g \underbrace{(g^{-1}Hg)}_{\in H} g^{-1} \\ &\subseteq gHg^{-1} \stackrel{②}{\subseteq} H \end{aligned}$$

We have proved that

$$gHg^{-1} = H \quad (*)$$

Then:

$$\begin{aligned} gH &= gH(g^{-1}g) = \underbrace{(gHg^{-1})}_{=H} g \stackrel{(*)}{=} \\ &= Hg \quad \text{for all } g \in G. \end{aligned}$$

So, H is normal



Example 5.5. Consider the general linear group $\overbrace{GL(2, \mathbb{R})}^G = \{A \in M_2(\mathbb{R}) : \det(A) \neq 0\}$. Then the subset $\underbrace{SL(2, \mathbb{R})}_H = \{A \in M_2(\mathbb{R}) : \det(A) = 1\}$ of G consisting on the matrices with determinant 1, called the *special linear group* is a normal subgroup of G .

Use the Subgroup Test and the properties of the determinant to prove that H is a subgroup of G .

$$H \stackrel{?}{\triangleleft} G.$$

Take $A \in G$, and show that $AHA^{-1} \stackrel{?}{\subseteq} H$.

For $B \in H$, we need to check that

$$\underline{ABA^{-1} \stackrel{?}{\in} H} \quad \left(\det(ABA^{-1}) \stackrel{?}{=} 1 \right)$$

$$\begin{aligned}
 \det(ABA^{-1}) &= \det(A) \det(B) \det(A^{-1}) \\
 &= \det(A) \underbrace{\det(B)}_1 \frac{1}{\det(A)} \\
 &= 1
 \end{aligned}$$

$$\Rightarrow ABA^{-1} \in H.$$

By the Normality Test, we have
that $H \triangleleft G$. $\blacksquare$